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Session based Song Recommendation Using Recurrent Neural Network

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Abstract. The music platforms contain large number of songs. The user is not likely to search the song which it may want to listen next. The recommendation system has solved this problem with collaborative filtering approach extensively. However, these recommendation systems are heavily relying on ratings information from the user. The absence of user information creates a new challenge in the music recommendation domain. The deep learning approaches found to be effective in sequence information learning. This investigation proposed a RNN based framework for session-based recommendation. This system predicts the next song based on the current ongoing session of songs. The proposed system uses improved session-parallel mini-batches and a ranking loss function to alter the basic RNN. This experiment shows that the relative improvement in ranking metric Precision@N, over the existing session-based methods.

Keywords: Recommendation System, collaborative filtering , Sequence aware recommendation system, Session-based Recommendation System, RNN

1 Introduction

Session-based item recommendation is a typical recommendation problem that may be seen in various domains such as e-commerce, news, hotel search, classified sites, music, and video recommendation. Because prior user history logs are sometimes unavailable in session-based settings (either because the user is new, not signed in, or not tracked), recommender systems must rely only on the user's activities in current sessions to deliver correct suggestions.

Most of these tasks were previously handled using very simple approaches such as collaborative filtering or content-based methods. Recurrent Neural Networks (RNNs) have evolved as powerful techniques for modelling sequential data in the deep learning. Speech recognition, machine translation, time series forecasting, and signal processing have all benefited from these models. RNNs have recently been used for the session-based recommendation setting in recommender systems, with improved results [20, 21]. RNNs have the benefit of being able to model the entire session of user interactions, as compared to conventional similarity-based[6] approaches for recommendation (clicks, views, etc.). RNNs may learn the 'theme' of a session by modelling the entire session, and hence

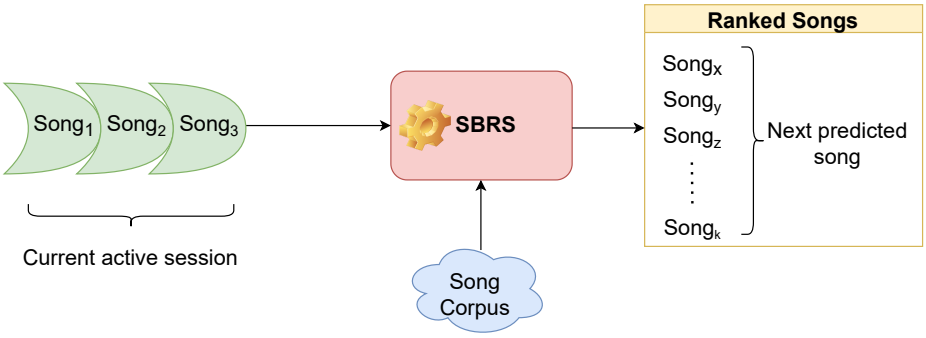


Fig. 1. Next song prediction model.

deliver recommendations with higher accuracy than previous approaches.

The task of session-based recommendation has been applied to RNNs. Ranking items according to the user preference is one of the main goal of recommendation system. It is less important to rank or score items at the bottom of the list (items that the user will not like), but it is critical to rank correctly the items at the top of the list (items that the user will like) (first 5 positions, 10 positions or 20 positions). To do this using machine learning, one must often employ learning to rank techniques, specifically ranking objectives and loss functions. Ranking loss functions, namely pairwise ranking loss functions, are used in contemporary session-based RNN techniques.

The deep learning algorithms propagates gradient across several layers. So, choosing a suitable ranking loss impacts majorly on the performance. in the case of RNNs 'back in time' over previous steps, to optimize model parameters, the quality of these gradients originating from the loss function has an impact on the quality of the optimization and the model parameters. The nature of recommendation task involves huge output space due to presence of large number of songs. It presents specific issues that must be considered while developing a good ranking loss function. We shall see that how this enormous output space issue is addressed is critical in reaching our goals.

The present investigation used ranking loss function in RNNs framework for session-based recommendation system. The BPR-max loss function is one of the popular ranking loss function that has been improve the RNN framework performance. The experiments have been done on 30Music dataset indicates a considerable boost in the recommendation results as assessed by Precision. The proposed framework shows the promising results over traditional recommendation systems.

The rest of the paper is organized as follows. Section 2, discusses about the existing works in this domain. The section 3 describes the proposed methodology of the SBRSs. In Section 4, the detailed statistical description of the dataset,

evaluation metric and comparative result analysis has been discussed. Finally, Section 5, concludes the work carried out in this paper.

2 Related Work

Most of the approaches proposed in the literature for session-based recommendation involve some form of sequence learning. Early versions of these techniques were based on the idea of frequent sequential patterns and Markov chain. Early techniques were used to predict the online behaviour of users. They were also used to predict users' recommendations in various e-commerce and music domain areas [2]. Much of the research in recommendation systems has been on models that operate when a user identification is available, and a clear user profile can be created. As per literature, most of the research carried out in this field are based on the models such as matrix factorization [3], frequent pattern mining [4], and nearest neighbourhood [5]. The item-to-item recommendation technique [6, 5] is one of the most common ways used in the session-based recommender system and a logical solution to the issue of a non-existent user profile. Items that are frequently clicked together in sessions are considered to be similar under this configuration, and an item-to-item similarity matrix is pre-computed from the available session data. During the session, this similarity matrix is simply leveraged to suggest the most similar products to the user that has been recently clicked. Despite its simplicity, it has been proved to be a successful and commonly used technique. While efficient, these approaches only consider the user's most recent click, thereby ignoring information from previous clicks.

Frequent pattern mining (FPM) [4] is one of the commonly used technique for SBRS, but the computational requirements becomes challenging. Also, finding suitable algorithm parameters can be challenging. In some cases, frequent item sequences do not lead to better predictions. Mobasher et al. [7] introduced one of the very first session-based methods based on frequent pattern mining for recommending the next web page to visit.

Markov Decision Processes(MDP) [8] are well suited for modeling sequential stochastic decision problems. It is defined as a four tuple $\langle S,\ A,\ Rwd,\ tr \rangle$, where S denotes the finite set of states, A denotes a set of actions (actions can be recommendations in case of recommender system), Rwd denotes reward function, and tr denotes the state-transition function. Shani et al. [8] proposed an MDP-based SBRS in the area of e-commerce and demonstrated the value of SBRS from a business perspective. Garcin et al. [9, 10] proposed an MDP-based SBRS that was applied in the news domain to recommend personalized news. Tavakol et al. [11] used a contextual session-based approach and provided a sequential framework for detecting the user's objective (the session's subject) using factored Markov decision processes (fMDPs). Along with MDPs, some other side information has been used on a private e-commerce dataset from a European fashion retailer. Jannach et al. [12] wanted to see how successful it is to use and combine long-term models with short-term adaption techniques. The findings show that preserving visitors' short-term content-based and recency-based profiles can lead

to considerable accuracy gains. To make sequential recommendations, Markov chain models assume that the user's future behaviour only depends on the very last behaviour or last few behaviours. Transition probability between items is evaluated in order to recommend the next item. The transition matrix provides the probability value of recommending an item by the user after consuming a sequence of items. The main drawback of applying Markov chains in the field of session-based recommendation systems is that when trying to capture all the possible sequences of user preferences becomes unmanageable.

Hariri et al. [13] proposed the nearest neighbors-based SBRS in order to generate a context-aware music playlist. More complex approaches based on Latent Markov Embedding were proposed for playlist generation by Chen et al. [14]. However, the nearest neighbors based technique and its modified variants have proved to be highly competitive against today's modern neural network methods [15]. The extended General Factorization framework (GFF) [19] also has the capability to use session data for recommendations and can be achieved by modeling a session by the sum of its event.

Hidasi et al. [17] presented the possibly first deep learning-based SBRS popularly known as gru4rec for predicting the imminent next item click event. They employed the YOOCHOOSE[1] dataset [18] for assessment and used an item-based nearest-Neighbors approach as one of their baselines for comparing the results. This model was continually improved later on by applying some different optimization objectives [19, 20]. In terms of prediction accuracy, they discovered that their neural network-based approach outperforms this methodology. Following their method and the increasing interest in neural networks, many other ideas using deep learning approaches to solve session-based recommendations have been deployed. It has been observed that the progress in the field of session-based recommendation seems still limited.

3 Proposed Method

The session based recommendation system is based upon the session sequences. The present investigation problem uses the sequence of songs. The RNN is well suitable model for the sequential data learning. The RNN model has been widely used in many different domains. In session based recommendation system, the RNN is capable to learn the sequential patterns of the session sequence. The fig. 2 shows the architecture of predicting the next song based on the previous session of songs sequence.

Problem definition: This work focuses on "next item recommendation" in recommendation system. The term 'item' here is referred as a song. We focus on the prediction problem here, which asks how effectively we can predict the next song the user desires to listen given a series of songs.

Definition of Next Song Recommendation A query q is a collection of songs in a session. Given a query q, the algorithm ranks all candidate songs and

[1] https://www.kaggle.com/chadgostopp/recsys-challenge-2015

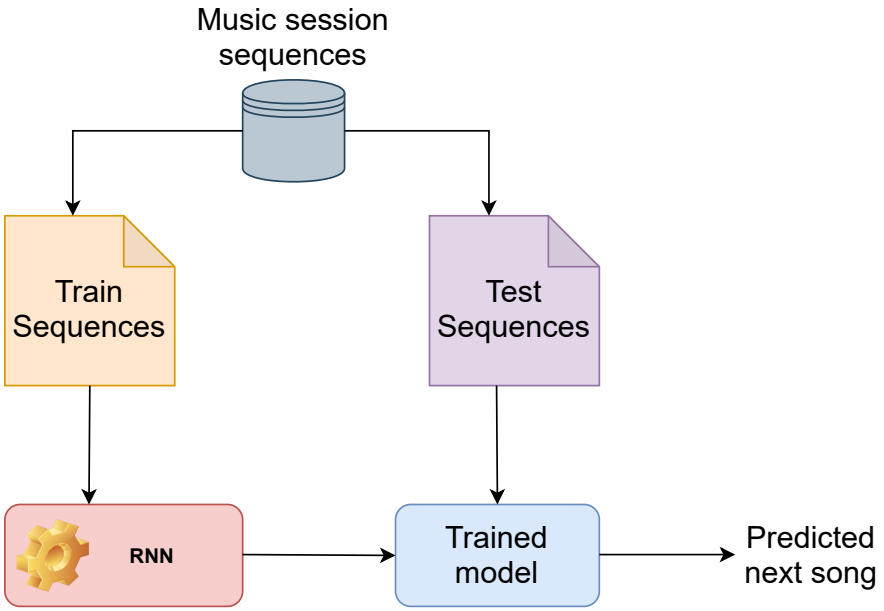


Fig. 2. Proposed framework for song prediction model using RNN.

recommends the top-k songs most likely to appear following the query q songs. Given a few songs, we predict the best song that may potentially come to the next of them, as seen in Figure 3.

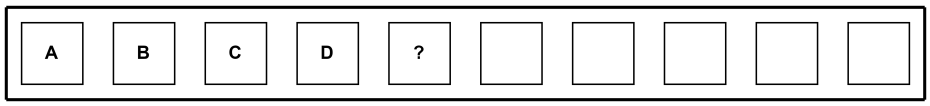


Fig. 3. Next song prediction problem.

The listened song log data is a valuable information and considered as an implicit feedback information. This requires no effort on user's side to collect their preferences and privacy concerns much. The proposed method predicts the next song based on previous sessions listened songs. Since, a person state of mind may vary according to the situation, time, activity, and location the current ongoing session is the most valuable information.

In session-based Recommendation, when a user enters a website and clicked on the first item is considered as the initial input of the Recurrent Neural Networks. After first click, each click of the user will produce an output as a recommendation and that output will be dependent on all the previous clicks. In order to encode the categorical data where features do not have any specific or-

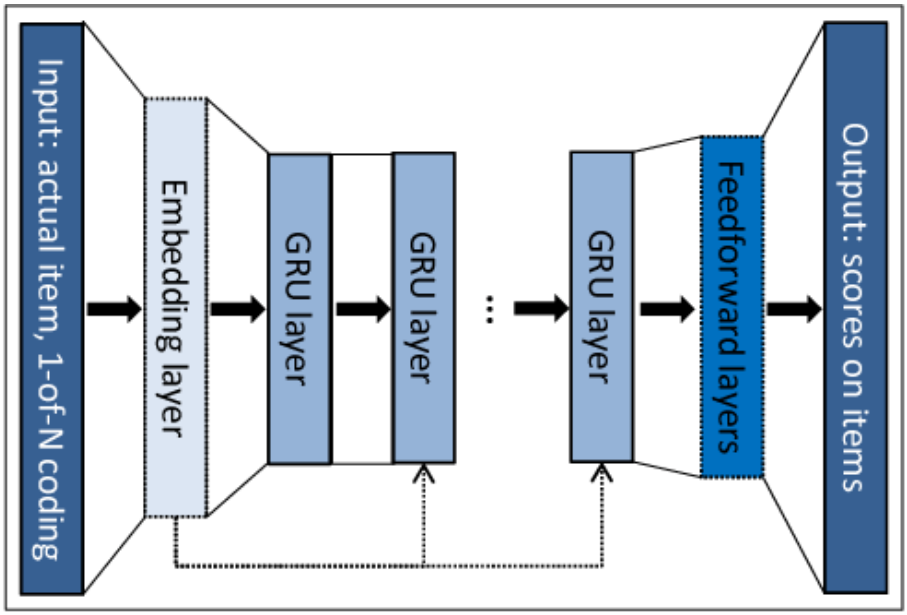


Fig. 4. Architecture of modified RNN used for song recommendation [17].

der 1-of-N encoding will be a good solution. In 1-of-N encoding, every song has been mapped as a binary variable. The 1-of-N encoding of the current session has been provided as an input to the RNN model. The term 'N' represents the total number of unique songs present in the song corpus. If the associated song is present in the current session, then the corresponding value will be 1, otherwise 0 in the 1-of-N encoding. The network's input is a single song that is 1-of-N encoded in a vector that represents the entire song space, and the network's output is a vector of similar shape that should yield a ranking distribution for the next song. In the present RNN model, the number of RNN layers are 20. The batch size and learning rate for the training of the RNN model has been kept as 16 and 0.1 respectively. The momentum of the model has been set to 0.1 and to avoid the overfitting of the RNN model, the dropout has been set as 0.1.

In order to efficiently train the proposed model session parallel mini batches, sampling technique for getting output and a BPR-max ranking loss function is used.

4 Result and discussion

In this section, details of the datasets used for evaluating the SBRSs based model is given as well as the pre-processing steps have been described. After that, the findings of our offline evaluation are presented and some important conclusion from the results have been drawn.

4.1 Dataset Description

30Music[2][1] dataset is a real-world publicly available collection of playlists and listening data. LastFM API was used to retrieve the data from internet radio stations. In this dataset each sample consists of four attributes i.e., *session_id*, *timestamp*, *item_id*, and *user_id*. In the pre-processing phase, all the sessions having only one event were removed.

Table 1. 30Music dataset characteristics.

Number of Sessions	2,764,474
Number of Events	31,351,954
Number of Users	4,106,341
Number of Items	210,633
Minimum Session Length	1
Maximum Session Length	198
Average Session Length	11
Minimum Number of Session Per User	1
Maximum Number of Session Per User	38
Average Number of Session Per Users	3.70

4.2 Dataset Splitting Approach

It should be emphasized that just a few research have addressed their data splitting techniques for model training, validation, and testing. It can be described by two primary types on the basis of whether the user information has been incorporated in the model creation or not. This has been examined on the basis of the public source codes of the relevant authors. Dataset can be split into training and testing in various ways. It depends on whether the user information has been incorporated in the model creation or not. Whenever *user_id* is presented in the dataset, it is better to split the dataset in such a way that every user's last session is kept in testing set and all the remaining session is kept in the train set. Whenever *user_id* is not present in the dataset, consider the most recent n days of user's sessions as the test set and remaining as the train set. In the present investigation, all the dataset contains *user_id*. So, in order to capture the personalized behavior of the users, this investigation follows the first approach. A user having n sessions have been split into training and testing set in such a way that first $(n-1)$ sessions will be in the training set and the last session (n^{th} session) of every user has been kept in the testing set.

Different ways of data filtering techniques lead to different data scenarios that would result in varying performance. Splitting of the dataset into train set and test set is done in such a way that the last session of every user is taken as a test set whereas all the remaining sessions are taken as a train set.

[2] http://recsys.deib.polimi.it/datasets/

4.3 Evaluation Metrics

Recommender systems can recommend a list of items at any point of time and the user might pick first few items from that list. Therefore, the primary evaluation metrics used in this work are Precision@N, Recall@N, and MRR@N. It is described in terms of recommendation systems as follows. Precision gives the number of items recommended which are relevant out of the total number of items recommended and could be defined by the Eq. 1. Recall gives the number of items recommended which are relevant out of the total number of items that are relevant and could be defined by the Eq. 2. Mean Reciprocal Rank (MRR) metric finds the rank of the first relevant item in the recommended list and then takes the reciprocal of the rank, the average across all the queries; the measure is known as Mean reciprocal rank. It is defined by the Eq. 3.

$$Precision = \frac{Number\ of\ items\ recommended\ which\ are\ relevant}{Number\ of\ items\ recommended} \tag{1}$$

$$Recall = \frac{Number\ of\ items\ recommended\ which\ are\ relevant}{Number\ of\ relevant\ items} \tag{2}$$

$$MRR = \frac{1}{|Q|}\sum_{i=1}^{|Q|}\frac{1}{ranK_i} \tag{3}$$

Table 2. Performance comparison at recommendation length of 5.

Model	Precision@5	Recall@5	MRR@5
POP	0.003	0.018	0.009
IKNN	0.076	**0.380**	**0.260**
SKNN	0.041	0.205	0.098
V-SKNN	0.195	0.137	0.088
RNN	**0.095**	0.073	0.223

Table 3. Performance comparison at recommendation length of 10.

Model	Precision@10	Recall@10	MRR@10
POP	0.003	0.030	0.010
IKNN	0.045	**0.454**	**0.270**
SKNN	0.028	0.285	0.010
V-SKNN	0.159	0.195	0.104
RNN	**0.064**	0.087	0.226

Table 4. Performance comparison at recommendation length of 20.

Model	Precision@20	Recall@20	MRR@20
POP	0.002	0.051	0.009
IKNN	0.025	**0.506**	**0.274**
SKNN	0.015	0.307	0.111
V-SKNN	0.112	0.244	0.110
RNN	**0.042**	0.100	0.228

4.4 Comparative Result

The result achieved by RNN model and three other models POP, IKNN, and SKNN on the 30Music dataset has been presented in the Table 2 to 4. To compare the performance of various methods on 30Music dataset, three metrics, Precision@N (P@N), Recall@N (R@N), and MRR@N, have been used, where the value of N is set to 5, 10,and 20. For all the metrics mentioned above, greater value implies better performance. The greatest value is written in bold across all approaches.

Table 2, 3 and 4 show that when the recommendation length is 5, RNN gives better results in terms of precision where as in terms of recall and MRR IKNN gives better result.

5 Conclusion and Future Work

Being able to predict the short-term interest of a user in an online session is a very relevant problem in practice. In this study, we have compared various computational and algorithmic approaches for session-based song recommendation. In this study we have used RNN to learn over session sequences in the training sequences set to predict the next song. By observing the above result, it can be seen that the recommendation quality changes when the length of the recommendation list varies.

SBRS generally ignores the user's long term preference that can be captured by the collaborative filtering based methods, so in future we can develop a model which can capture both short term preferences as well as long term preferences. Along with session context, features of items of the session can be incorporated to improve the quality of recommendation.

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